# A PROPER SCHEME OF DIMENSION AT MOST 1 OVER A FIELD IS H-PROJECTIVE

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Allowed standard prerequisites (editorial).*

- Dim-1 separated finite type over a field is H-quasi-projective. delivered companion lemma-dim-1-quasi-projective. First input.
- Proper schemes have closed images in separated schemes. Morphisms, Lemma morphisms-lemma-image-proper-scheme-closed. Second input.

*Source reference.* Original lemma 0A25: sources/stacks-chow/tags.txt, source lines 12693–12693. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $X$ be a proper scheme over a field $k$. If $\dim(X) \leq 1$ then $X$ is H-projective over $k$.*

*Proof.* By Lemma 0A25 we see that $X$ is a locally closed subscheme of $\mathbf{P}^n_k$ for some field $k$. Since $X$ is proper over $k$ it follows that $X$ is a closed subscheme of $\mathbf{P}^n_k$ (Morphisms, Lemma 01W6). □

*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.